Problem

Title

Strong Number Checker (Factorial Digit Sum)

Topics

Number Mathematics, Factorial, Digit Extraction, Nested Loops, Conditional Statements

Difficulty Level

Easy

Problem Detailed Statement (Scenario Based)

In a mathematical coding challenge, students are introduced to another special category of numbers called **Strong Numbers**.

A **Strong Number** is defined as a number in which the sum of the factorial of its digits is equal to the number itself.

For example:

- 145
  (1! + 4! + 5!)
  = 1 + 24 + 120
  = 145 ✓
- 2
  (2! = 2) ✓

However,

- 123
  (1! + 2! + 3! = 1 + 2 + 6 = 9) ✖ (Not equal to 123)

Now, the trainer provides a positive integer **N**. Students must determine whether the number is a Strong Number or not.

To solve this problem, students must:

1. Extract each digit of the number using loop logic.
2. Compute the factorial of each digit using a nested loop.
3. Maintain a running sum of all factorial values.

4. Compare the final sum with the original number.
5. Print appropriate output.

Restrictions for deeper learning:

- Do not use built-in factorial libraries.
- Do not convert the number into a string for digit access.
- Use proper looping and conditional structures.

This problem strengthens:

- Nested loop implementation
- Mathematical computation
- Logical thinking
- Factorial calculation
- Edge case handling (single digit numbers)

Input Format

A single integer:

`N`

Constraints:

`0 ≤ N ≤ 10^7`

Output Format

- Print `"Strong Number"` if N is a Strong Number.
- Otherwise print `"Not Strong Number"`.

Examples

Example 1

Input:

```
145
```

Output:

```
Strong Number
```

Example 2

Input:

```
2
```

Output:

```
Strong Number
```

Example 3

Input:

```
123
```

Output:

```
Not Strong Number
```

Example 4

Input:

```
40585
```

Output:

```
Strong Number
```
